

WHAT CHANGES MATTER? FUNCTIONAL SUPPORT OF FINE-TUNING SHIFTS IN SPEECH RECOGNITION

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ABSTRACT

Fine-tuning improves speech recognition by changing an encoder’s representation, but it is unclear which parts of that change the adapted recognizer actually uses. We study this question with a controlled intervention on a Wav2Vec2-Large-960h recognizer. We define the input-conditioned representation shift as the difference between fine-tuned and pre-trained final-layer representations, estimate coordinate utility with a first-order Taylor criterion derived from the fine-tuned CTC loss, and intervene while keeping the original CTC readout fixed. On the official L2-ARCTIC UT8 Fold 0 split, the complete shift reaches 10.499% test WER, whereas removing it from the adapted readout gives 118.912%. Retaining the top half of the 1024 shift coordinates gives 11.713%, preserving 98.9% of the adaptation gain. The result shows that the functional effect of fine-tuning is concentrated in a small subset of representation changes.

Index Terms— ASR, SSL, CTC, selection

1. INTRODUCTION

Self-supervised learning (SSL) has become a central paradigm for speech representation learning, enabling large encoders to exploit unlabeled audio and adapt to downstream tasks with limited supervision [1, 2, 3, 4, 5, 6]. Fine-tuning such models is now widely used to adapt general-purpose speech representations to challenging domains, including child, accented, noisy, and other mismatched speech [7, 8, 9, 10]. While these studies demonstrate the effectiveness of adaptation through endpoint recognition performance, considerably less is known about *which internal representation changes actually support the resulting recognition behavior*.

A growing body of representation analysis has examined what SSL speech models encode and how this information evolves across layers. Acoustic, phonetic, speaker, lexical, and higher-level linguistic information has been localized using probing and representation-similarity analyses, with ASR fine-tuning producing particularly strong changes in later encoder layers [11, 12, 13, 14, 15]. Related work has further shown that manipulating the representation space through self-supervised rewiring or representation-focused fine-tuning can improve downstream speech model-

ing [16, 17]. These analyses characterize *what is represented* and *where representations change*, but do not establish which fine-tuning-induced changes are functionally used by the adapted recognizer.

In parallel, differences between pretrained and fine-tuned models have emerged as meaningful objects of study. Weight-space interpolation and task arithmetic show that fine-tuning-induced parameter differences encode reusable adaptation information [18, 19]; this idea has subsequently been applied to ASR through task-vector algebra and model merging across domains [20, 21, 22]. At the representation level, *Mind the Shift* shows that the difference between fine-tuned and pre-trained SSL embeddings—the *Delta embedding*—contains task-relevant information that improves ASR when fused with another SSL representation [23]. Recent representation-space interventions similarly suggest that speech attributes such as accent can be localized and manipulated directly in hidden activations [24]. However, showing that a complete representation shift is useful does not reveal how its usefulness is distributed internally. *Does an adapted recognizer rely broadly on the entire fine-tuning shift, or only on particular components of it?*

Answering this question requires distinguishing the *size* of a representation change from its effect on the learned decision rule. Magnitude alone need not reflect functional importance: sensitivity- and Taylor-based criteria have long shown that task loss can provide a more direct measure of the effect of removing model components [25, 26, 27], while masking-based interventions have been used to test the faithfulness of explanations in audio models [28]. Motivated by this distinction, we introduce **Decision-Aligned Delta Selection (DADS)**. For an utterance x , we define the fine-tuning shift as $\Delta(x) = E_{\text{ft}}(x) - E_{\text{pt}}(x)$ and rank its coordinates using a first-order Taylor score that combines each displacement with the sensitivity of the fine-tuned CTC loss. We then retain or remove selected Delta coordinates while keeping the original fine-tuned CTC head and decoding procedure fixed. Thus, no downstream retraining can compensate for the intervention: retention tests which changes are sufficient for recognition, while deletion tests which changes the adapted recognizer critically depends on.

Experiments on three official L2-ARCTIC UT8 folds reveal a strongly non-uniform dependence on the fine-tuning

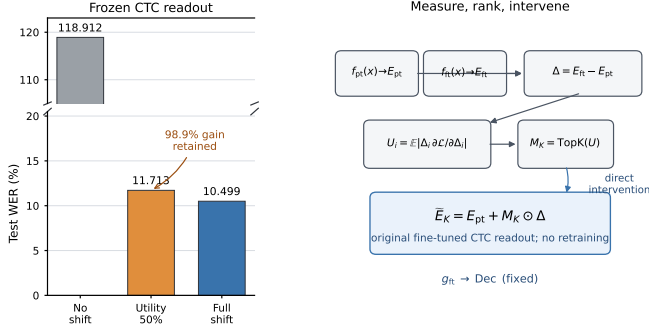


Fig. 1. DADS measures and intervenes on the fine-tuning shift. Paired representations yield $\Delta = E_{ft} - E_{pt}$; Taylor utility ranks its coordinates; a retained mask is supplied to the original fine-tuned CTC readout. The left panel shows the official Fold 0 intervention at the 50% budget.

shift. On Fold 0, the complete shift achieves 10.499% WER; retaining 75% of DADS-ranked coordinates gives 10.585%, while retaining half gives 11.713% compared with 16.638% for matched magnitude selection. More strikingly, removing the lowest-utility quarter leaves WER at 10.585%, whereas removing the highest-utility quarter raises it to 131.190%. The same ordering is reproduced on two additional folds. These results show that fine-tuning changes a speech representation broadly, but the adapted recognizer depends on those changes highly unevenly.

Our contributions are threefold: (1) we study how recognition-relevant information is distributed within fine-tuning-induced representation shifts; (2) we propose DADS and a frozen-head intervention for identifying decision-relevant Delta coordinates without retraining; and (3) we show across official L2-ARCTIC folds that large portions of the shift are comparatively dispensable, while a concentrated set of decision-aligned changes is critical to the adapted recognizer.

2. METHOD

Overview. We treat fine-tuning as a measurable intervention on representation space. For the same utterance, the pretrained and adapted encoders expose a paired shift; a binary gate then supplies selected coordinates of that shift to the already adapted readout. This turns functional support into a controlled counterfactual: the encoder, readout, decoding rule, and examples stay fixed while the supplied shift changes. Decision-Aligned Delta Selection (DADS) estimates the support with a one-shot Taylor utility and tests the estimate through complementary retention and deletion interventions. Figure 1 gives the complete measurement loop.

2.1. Fine-tuning shifts as paired representation changes

Let f_{pt} and f_{ft} denote pretrained and fine-tuned versions of the same encoder, evaluated on the same waveform and with the same preprocessing. For an utterance x , their final-layer representations are

$$E_{pt}(x) = f_{pt}(x), \quad E_{ft}(x) = f_{ft}(x), \quad (1)$$

where $E_{pt}(x), E_{ft}(x) \in \mathbb{R}^{T_x \times d}$. Because both representations are obtained from the same encoder architecture and the same input, their difference defines an input-conditioned fine-tuning shift,

$$\Delta(x) = E_{ft}(x) - E_{pt}(x). \quad (2)$$

Thus,

$$E_{ft}(x) = E_{pt}(x) + \Delta(x). \quad (3)$$

We study Δ in the encoder’s native representation basis and preserve its frame axis. The unit of selection is therefore a coordinate of the representation shift, not a parameter, a source layer, or a learned latent feature.

2.2. Frozen-readout interventions and functional support

Let g_{ft} denote the CTC readout learned by the fine-tuned recognizer, and let Dec denote the fixed greedy CTC decoding rule. We intervene on the representation shift while leaving this adapted decision rule unchanged. For a binary coordinate mask $m \in \{0, 1\}^d$, shared across time, define

$$\tilde{E}_m(x) = E_{pt}(x) + m \odot \Delta(x), \quad (4)$$

where m is broadcast over frames. A one keeps a shift coordinate; a zero removes it. The encoder and CTC readout remain fixed, so masking is a counterfactual on the supplied shift, not retraining of a smaller recognizer.

The two endpoints have a direct interpretation:

$$\tilde{E}_1(x) = E_{ft}(x), \quad \tilde{E}_0(x) = E_{pt}(x). \quad (5)$$

The zero-mask endpoint is therefore not a pretrained-ASR baseline. It is the pretrained representation evaluated by the frozen fine-tuned readout: the representation change is removed, but the decision rule that learned to use it is preserved.

Let $[d] = \{1, \dots, d\}$, let $S \subseteq [d]$ denote a subset of retained coordinates, and define its mask by $m_S(i) = \mathbb{1}[i \in S]$. We define the functional support of the shift as the smallest coordinate subset that preserves the behavior of the adapted recognizer. For a behavioral tolerance ϵ ,

$$S_\epsilon^* = \arg \min_{S \subseteq [d]} |S| \quad \text{s.t.} \quad \mathcal{B}(m_S) \leq \mathcal{B}(1) + \epsilon, \quad (6)$$

where $\mathcal{B}(m)$ denotes recognition error under Dec($g_{ft}(\tilde{E}_m)$) on a fixed evaluation set. The 2^d possible masks make exhaustive search impractical; DADS supplies a one-shot ordering whose sufficiency and necessity can be tested directly.

2.3. Decision-Aligned Delta Selection

We estimate coordinate importance from the local sensitivity of the actual fine-tuned CTC objective rather than from shift magnitude alone. For an utterance–transcript pair (x, y) , introduce a continuous coordinate gate $\alpha \in [0, 1]^d$ and define

$$\mathcal{L}_{x,y}(\alpha) = \mathcal{L}_{\text{CTC}}(g_{\text{ft}}(E_{\text{pt}}(x) + \alpha \odot \Delta(x)), y). \quad (7)$$

The full fine-tuned representation corresponds to $\alpha = \mathbf{1}$. The sensitivity of the loss to coordinate i at this endpoint is

$$\left. \frac{\partial \mathcal{L}_{x,y}}{\partial \alpha_i} \right|_{\alpha=\mathbf{1}} = \sum_{t=1}^{T_x} \Delta_{t,i}(x) \frac{\partial \mathcal{L}_{x,y}}{\partial \Delta_{t,i}(x)}. \quad (8)$$

A first-order removal therefore depends jointly on how much coordinate i changed during fine-tuning and how sensitive the adapted CTC objective is to that change. Magnitude alone captures only the former. For each frame and coordinate, define the local decision-aligned contribution

$$q_{t,i}(x, y) = \Delta_{t,i}(x) \frac{\partial \mathcal{L}_{x,y}}{\partial \Delta_{t,i}(x)}. \quad (9)$$

The global derivative in Eq. 8 can cancel across frames. DADS instead uses a non-cancelling framewise aggregation, preserving the local evidence that the CTC objective assigns to each encoder frame. Given a fixed utility-estimation subset $\mathcal{D}_u \subset \mathcal{D}_{\text{train}}$, we define

$$U_i = \frac{1}{N_u} \sum_{(x,y) \in \mathcal{D}_u} \sum_{t=1}^{T_x} |q_{t,i}(x, y)|, \quad N_u = \sum_{(x,y) \in \mathcal{D}_u} T_x. \quad (10)$$

This non-cancelling aggregation is related to the global coordinate-gate sensitivity through

$$\left| \frac{\partial \mathcal{L}_{x,y}}{\partial \alpha_i} \right| = \left| \sum_t q_{t,i} \right| \leq \sum_t |q_{t,i}|. \quad (11)$$

Thus, U_i is a first-order, decision-aligned saliency: it combines the size of the learned shift with the sensitivity of the true CTC objective to that shift. In implementation, the per-utterance CTC loss is normalized by transcript length before differentiation and padded frames are excluded. For the magnitude baseline, coordinates are ranked by the same frame-normalized aggregation of $|\Delta_{t,i}|$; random masks are sampled at the same retained budget.

The ranking is computed once from $\mathcal{D}_u$. No encoder parameters, CTC parameters, decoder parameters, or ranking scores are updated after coordinate selection.

2.4. Retention and deletion interventions

Let π denote the ordering of coordinates induced by U_i . We use the same ranking for two complementary interventions.

Table 1. Retention results with the frozen fine-tuned CTC readout on official UT8 Fold 0. Lower WER is better.

Condition	Coordinates	Test WER (%)
No shift	0/1024	118.912
Full shift	1024/1024	10.499
DADS, 75%	768/1024	10.585
DADS, 50%	512/1024	11.713
Magnitude, 50%	512/1024	16.638

Retention: testing sufficiency. For a budget K , let

$$S_K^{\text{keep}} = \{\pi_1, \dots, \pi_K\}, \quad (12)$$

and evaluate

$$g_{\text{ft}}(E_{\text{pt}}(x) + m_{S_K^{\text{keep}}} \odot \Delta(x)). \quad (13)$$

This is the sufficiency test: are the highest-utility coordinates enough to recover the behavior induced by the complete shift?

Deletion: testing necessity. We additionally begin from the complete shift and remove either the highest- or lowest-utility coordinates. For a deletion budget K ,

$$S_K^{\text{drop-high}} = [d] \setminus \{\pi_1, \dots, \pi_K\}, \quad (14)$$

$$S_K^{\text{drop-low}} = [d] \setminus \{\pi_{d-K+1}, \dots, \pi_d\}. \quad (15)$$

This is the necessity test: are the coordinates identified by DADS more necessary to the adapted decision rule than coordinates ranked as low utility?

All interventions use the same frozen CTC readout and the same decoding procedure. Consequently, any behavioral difference between masks is produced by the supplied representation shift rather than by downstream retraining or recovery.

3. RESULTS

We organize the results around three questions: how much of the shift is sufficient, which coordinates are necessary, and whether the pattern is reproduced across folds.

3.1. How much of the shift is sufficient?

We first compare the complete shift, the no-shift counterfactual, and matched-budget coordinate selection on official UT8 Fold 0. Table 1 shows that retaining only 512 of the 1024 shift coordinates gives 11.713% WER and preserves 98.9% of the full adaptation gain. At the same budget, magnitude selection gives 16.638%. Retaining 75% of the coordinates gives 10.585%, close to the full-shift result of 10.499%.

The large gap between the no-shift and full-shift endpoints establishes that the adapted readout uses the fine-tuning change. The near-full performance of the retained

Table 2. Necessity test under 25% deletion on official UT8 Fold 0.

Condition	Test WER (%)
Full shift	10.499
Drop highest-utility 25%	131.190
Drop lowest-utility 25%	10.585

masks establishes a separate result: a substantial part of that change is functionally redundant under the frozen decision rule. This is a behavioral redundancy result, not a claim that half of the encoder can be removed computationally.

3.2. Which coordinates are necessary?

Retention alone tests sufficiency. To test necessity, we start from the complete shift and remove either the highest- or lowest-utility quarter. Removing the highest-utility coordinates raises WER to 131.190%, whereas removing the lowest-utility coordinates gives 10.585%. The same ranking that finds a sufficient retained subset therefore distinguishes behaviorally critical coordinates from coordinates that can be removed with little effect. The matched Magnitude mask at $K = 512$ gives 16.638% WER, compared with 11.713% for DADS, showing why shift size alone is not an adequate selection criterion.

3.3. Does the finding transfer across folds?

We repeat the matched 50% comparison with independently trained replicas on two additional official folds. DADS reaches 16.011% and 15.579% WER on Folds 1 and 2, compared with 16.628% and 16.485% for magnitude selection. Dropping the top utility quarter is also consistently more damaging than dropping the bottom quarter. The ordering therefore does not depend on the single Fold 0 checkpoint, although the present evidence remains tied to this encoder family, representation basis, and frozen-readout protocol.

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